

Edge computing with AI-driven anomaly detection for resilient critical infrastructure in urban environments.

Public / Government sector × Critical infrastructure monitoring × Edge computing · OS072 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
57 is the world moving	58 can we play, can we win	HIGH 1 named in the evidence	€220m observed evidence · per year	LATER research and standards activi...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying AI-driven anomaly detection at the edge to make critical urban infrastructure—such as power grids, water systems, and transport networks—more resilient. It targets public sector operators and government agencies responsible for these systems, and it exists now because edge AI has matured enough to run on resource-constrained devices while research projects like RESCUE are validating the approach.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Observed: tendered contract value, annualised	€220m €220m – €482m	€220m €220m – €482m	€6.2m €6.2m – €13.5m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

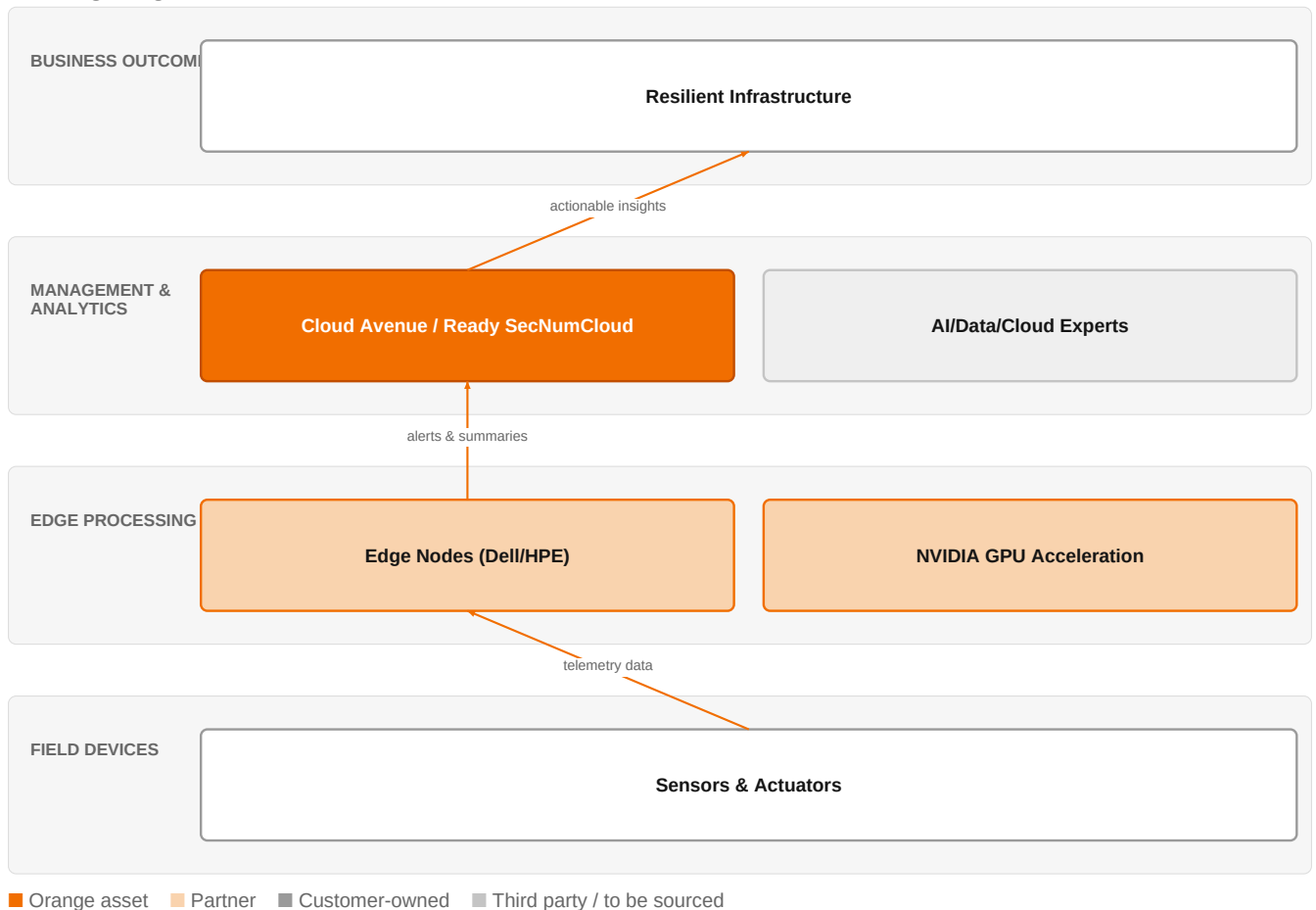
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€220m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	2.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 34 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Sovereign Edge AI for Critical Infrastructure



This picture shows how Orange combines sovereign cloud, edge processing, and AI expertise to deliver real-time anomaly detection for critical infrastructure.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Edge AI has evolved from experimental to practical: lightweight models like FA-YOLOv11 can now run on industrial edge devices with acceptable latency, and frameworks like Cloudburst AI orchestrate inference across multi-cloud and edge zones. Energy-aware inference for low-power IoT devices is also emerging, addressing the power constraints of urban sensors. The RESCUE project explicitly targets resilient edge systems for critical infrastructure, signaling a coordinated research push. However, the evidence is mostly academic; there is no strong commercial signal yet, and the topic is described as 'fading' in its lifecycle.

Sources: SIG-5B350B1A3B15 Journal of Organizatio 2026-05-21; SIG-3772E07403F3 openalex.org 2026-01-02; SIG-D4C81A60CDDC openalex.org 2025-10-17; SIG-F292FB8C24B6 cordis.europa.eu 2025-09-01

Who buys, and why now

The buyer is typically the chief technology officer or head of infrastructure at a public utility, city government, or national agency responsible for critical infrastructure. The pain is operational: legacy monitoring is reactive, cloud-based analysis introduces latency that can miss fast-moving faults, and cyber threats are increasingly sophisticated. The trigger is often a near-miss incident—a brief outage, a security breach, or a regulatory audit that exposes gaps in real-time visibility. They need to detect anomalies at the source, before they cascade.

Why it is hot now — every claim bound to a dated source

- The RESCUE project aims to strengthen the resilience, security, and adaptability of critical infrastructure systems through advanced edge computing, AI-driven anomaly detection, and robust communication networks. [SIG-F292FB8C24B6]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a sovereign edge AI solution: starting with a certified cloud foundation (Cloud Avenue / Ready SecNumCloud) to host the management plane, then deploying edge nodes using partner hardware from Dell or HPE, with NVIDIA for GPU acceleration. Orange's AI/Data/Cloud experts would build and tune the anomaly detection models, while Cybersecurity experts ensure the edge-to-cloud path is secure. The engagement would follow a phased approach: assess the customer's infrastructure, pilot on a single site, then scale across the estate.

Why Orange can win

Orange's differentiators are its sovereign certifications (SecNumCloud, ISO 27001) and its ability to integrate edge, cloud, and security—assets that competitors like AWS or Accenture cannot fully match for public sector buyers. The reference of Grand Paris Sud demonstrates experience with urban-scale deployments. However, the asset list is thin on specific edge AI offerings; Orange would need to rely on partner technology and its own services to assemble a competitive solution.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Cloud Avenue / Ready SecNumCloud	offer	L1	offer provides the technology in a matching domain	unconfirmed
Orange Drone Guardian	offer	L1	offer addresses the use case but not this technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Dell	partner	L2	partner provides the required technology	unconfirmed
HPE	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
CISPE	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Grand Paris Sud	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

HIGH competitive intensity (score 8.3; 1 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

AWS is strong in cloud and edge infrastructure, and is also an Orange partner, so the customer may compare against AWS's edge services. Deutsche Telekom/T-Systems has a strong public sector presence and edge offerings. Lumen and NTT DATA also sell edge computing, with Lumen focusing on cybersecurity. Schneider Electric brings deep OT expertise, while HPE (also a partner) and Nokia offer network-centric edge solutions. Accenture is active in the public sector and could position as an integrator. The customer will likely compare on price, sovereignty, and integration complexity.

Sources: SIG-701F806B6A7A International Journal 2025-07-30

Competitor	Category	Position here	Basis	Evidence
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AWS	Hyperscaler	sells Edge computing; competes in Cloud — also an Orange partner	evidenced	SIG-701F806B6A7 A
Deutsche Telekom / T-Systems	Telecom operator peer	sells Edge computing; active in Public / Government sector; competes in Cloud, Cybersecurity	structural	—
Lumen Technologies	Telecom operator peer	sells Edge computing; competes in Cybersecurity	structural	—
NTT DATA	Global systems integrator	sells Edge computing; competes in Cloud	structural	—
Schneider Electric	Industrial / OT specialist	sells Edge computing	structural	—
HPE (Aruba, Athonet)	Network equipment vendor	sells Edge computing — also an Orange partner	structural	—
Nokia	Network equipment vendor	sells Edge computing	structural	—
Accenture	Global systems integrator	active in Public / Government sector; competes in Cloud	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is the current latency between anomaly detection and response in your critical infrastructure monitoring?
- 2 Have you experienced a recent incident where cloud-based analysis was too slow to prevent or mitigate an outage?
- 3 What are your data sovereignty requirements for edge-collected data from critical infrastructure?
- 4 Do you have existing edge devices or gateways deployed at your sites that could host AI models?
- 5 Which of your critical assets are most vulnerable to cyber-physical attacks, and how is that risk currently managed?
- 6 What is your budget cycle for infrastructure modernization, and is there a specific project where edge AI could be piloted?

Objections you will meet

They will say	You can answer
Edge AI is too immature for our critical infrastructure; we can't risk downtime.	That's a fair concern. We're not proposing to replace your existing monitoring overnight. We would start with a pilot on a non-critical asset, prove the reliability, and then scale. Our approach is designed to add resilience, not introduce risk.
We already have a cloud provider (like AWS) that offers edge services.	True, hyperscalers have edge offerings. But for public sector critical infrastructure, sovereignty and certification often matter more. Our SecNumCloud-certified cloud and our ability to integrate with your existing systems give you control that a hyperscaler may not provide.
This sounds expensive—we have budget constraints.	We understand budget pressures. That's why we propose a phased approach: start with a small pilot to demonstrate value, then scale based on measurable outcomes. We can also explore co-funding through EU research programs, as seen in projects like RESCUE.

Risks and unknowns

The main risk is that the market is not yet ready: the evidence is mostly academic, and the topic is flagged as 'fading,' suggesting limited commercial traction. Customers may be hesitant to adopt edge AI for critical infrastructure due to security concerns or lack of in-house skills. Unknowns include the actual performance of edge AI on legacy infrastructure and the willingness of public sector buyers to invest in a technology that is still maturing.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a public-sector tender on critical infrastructure monitoring, leveraging the RESCUE project's findings, Orange's ISO 27001 and SecNumCloud certifications, and partnerships with NVIDIA, Dell, and HPE to differentiate the solution.

Sales	In a conversation with a European city's public safety or urban infrastructure director, say: 'Orange combines its SecNumCloud-certified Cloud Avenue with AI-driven anomaly detection and edge computing expertise to help cities like Grand Paris Sud strengthen the resilience of critical infrastructure.'
Strategist / Innovator	Commission a deep-dive brief on the RESCUE project's edge computing and AI anomaly detection architecture, mapping how Orange's AI/Data/Cloud experts and cybersecurity experts can contribute to a reference architecture for urban critical infrastructure resilience.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-5B350B1A3B15	2026-05-21	Journal of Organizational...	1	Latency Trade-Offs of a Lightweight YOLOv11 Detection Algorithm on Edge Computing Nodes for Fac...
SIG-3772E07403F3	2026-01-02	openalex.org	1	Cloudburst AI: Orchestrating SRE-Driven AI Inference Across Multi-Cloud and Edge Zones
SIG-04BD044A1960	2025-11-18	openalex.org	1	Cognitive Edge Computing: A Comprehensive Survey on Optimizing Large Models and AI Agents for P...
SIG-D4C81A60CDDC	2025-10-17	openalex.org	1	An Energy-Aware Generative AI Edge Inference Framework for Low-Power IoT Devices
SIG-F48074B55B20	2025-10-16	openalex.org	1	Implementation of edge AI for early fault detection in IoT networks: evaluation of performance ...
SIG-D031822F4DA6	2025-09-09	openalex.org	1	Closed-Loop Network Control for Industrial Edge Computing: A Telemetry-Driven and AI-Based Appr...
SIG-F292FB8C24B6	2025-09-01	cordis.europa.eu	1	Resilient Edge Systems for Critical infrastructure and Urban Environments
SIG-B7834050E076	2025-08-26	openalex.org	1	A Case Study of Environmental Footprints for Generative AI Inference: Cloud versus Edge
SIG-701F806B6A7A	2025-07-30	International Journal of ...	1	Edge Computing in Networks: Reducing Latency Using AI-Driven Edge Computing Strategies
SIG-C125FF99E68A	2025-02-26	The Role of Network Secur...	1	Smart Cities with 5G and Edge Computing in 2030
SIG-D3644CC23299	2025-01-01	cordis.europa.eu	1	Collaborative Open Platform for PILOTing services across emerging smart IoT and Edge computing ...
SIG-575FE627E0DA	2026-07-11	arxiv.org	3	CSI-Assisted Edge SLAM Testbed Platform for 5G Connected Unmanned Autonomous Vehicles
SIG-565963A66A79	2026-06-01	arxiv.org	3	IstGPT: LLM-based Anomaly Detection for Spatial-Temporal Graph in Industrial Systems

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:44+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.